

UNIVERSAL TRANSLATOR

Step-by-Step Roadmap to Make the App Feel Truly Universal

Build the translator into a live interpreter that detects language, routes speakers, repairs uncertainty, remembers context, and keeps working when the network is weak.

PRIMARY GOAL

Tap once, speak naturally, translate and voice continuously.

BRAIN ROLE

Use AI for decisions, confidence, repair, and context.

MOBILE STANDARD

Clean live interpreter screen with no setup friction.

Build Steps

1

True Streaming STT

Goal: speech while talking becomes live words immediately.

- Add a provider such as Deepgram, AssemblyAI, Google Speech-to-Text streaming, or Azure Speech.
- Stream microphone audio continuously instead of waiting for final clips.
- Use backend Whisper/browser recognition only as fallback.

2

Language Auto-Detection

Goal: user speaks English, Spanish, Haitian Creole, or another language and the app chooses the source language.

- Detect language from the live transcript.
- Update the speaker profile with the detected language.
- Switch translation direction automatically.

3

Speaker Routing

Goal: the app knows who is talking and what language each person uses.

- Track each phone/device as a speaker identity.
- Bind each speaker to a default language and target language.
- Route every transcript through the correct speaker direction.

4

AI Comm Brain Decisions

Goal: use the AI brain for conversation intelligence, not raw speech speed.

- Handle unclear meaning, interruptions, speaker changes, names, numbers, and emotional tone.
- Decide whether to translate immediately or ask for clarification.
- Use context from the current conversation before choosing the final phrasing.

5

Confidence Repair

Goal: high confidence translates instantly; low confidence asks for repair.

- Score every transcript and translation.
- Ask for exact repairs like: "Please repeat the name" or "Confirm the number."
- Ask meaning repair when needed: "Did you mean this or that?"

6

Voice Output Polish

Goal: translated voice feels fast, natural, and consistent.

- Stream TTS chunks instead of waiting for one full audio file.
- Cache common phrases and warm the voice engine before speech starts.
- Keep a stable translated voice per speaker.

7

Weak-Network Fallbacks**Goal: the app keeps working in bad network conditions.**

- Best mode: live streaming STT plus live TTS.
- Backup modes: browser speech recognition, backend Whisper partials, then text-only translation.
- Show the active mode clearly in the mobile UI.

8

Conversation Memory**Goal: the app remembers context during the conversation.**

- Track names, places, numbers, speaker languages, recent turns, and corrections.
- Feed memory into translation refinement and repair decisions.
- Keep short-term memory fast and privacy-safe.

9

Mobile UX Polish**Goal: the phone screen shows only what matters.**

- Keep language direction, big mic, live transcript, live translation, speaker name, and live status bar.
- Hide diagnostics unless debug mode is intentionally opened.
- Make one tap start continuous live interpreter mode, and tap again stop it.

Best Order to Build**1 True streaming STT****2 Language auto-detection****3 Speaker routing****4 AI comm brain decisions****5 Confidence repair****6 Streaming TTS polish****7 Offline/weak-network fallback****8 Mobile UX polish****9 Real-world testing****Real Conversation Tests****English to Spanish****Spanish to English****English to Haitian Creole****Noisy room****Two people talking****Interruptions****Names and numbers****Weak internet**